Solution

Forces on block A:

(i) Downward gravitational force exerted by the Earth ($m_A g$)
(ii) Upward normal force exerted by block B (N_B)

The free body diagram for block A is as shown in the following picture.

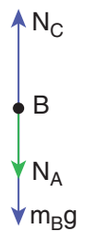
Force on block A



Forces on block B :

(i) Downward gravitational force exerted by Earth ($m_B g$)
(ii) Downward force exerted by block A (N_A)
(iii) Upward normal force exerted by block C (N_C)

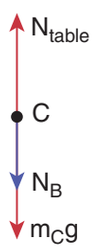
Force on block B



Forces onblock C:

(i) Downward gravitational force exerted by Earth ($m_C g$)
(ii) Downward force exerted by block B (N_B)
(iii) Upward force exerted by the table (N_{table})

Force on block C



EXAMPLE 3.12

Consider a horse attached to the cart which is initially at rest. If the horse starts walking forward, the cart also accelerates in the forward direction. If the horse pulls the cart with force F_h in forward direction, then according to Newton's third law, the cart also pulls the horse by equivalent opposite force $F_c = F_h$ in backward direction. Then total force on 'cart+horse' is zero. Why is it then the 'cart+horse' accelerates and moves forward?

Solution

This paradox arises due to wrong application of Newton's second and third laws. Before applying Newton's laws, we should decide 'what is the system?'. Once we identify the 'system', then it is possible to identify all the forces acting on the system. We should not consider the force exerted by the system. If there is an unbalanced force acting on the system, then it should have acceleration in the direction of the resultant force. By following these steps we will analyse the horse and cart motion.

If we decide on the cart+horse as a 'system', then we should not consider the force exerted by the horse on the cart or the force exerted by cart on the horse. Both are internal forces acting on each other. According to Newton's third law, total internal force acting on the system is zero and it cannot accelerate the system. The acceleration of the system is caused by some external force. In this case, the force exerted by the road on the system is the external force acting on the system. It is wrong to conclude that the total force acting on the system (cart+horse) is zero without including all the forces acting on the system. The road is pushing the horse

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